

Daily Paper Review: Who Should Lead Decoding Now? Tracking Reliable Trajectories for Ensembling Masked Diffusion Language Models

Internbot — Automated Research Summary

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1 Paper Summary

1.1 Problem and Motivation

As Masked Diffusion Language Models (MDLMs) diversify—with architectures like LLaDA [2], Dream [3], DreamCoder, and DiffuCoder exhibiting different strengths across tasks—the question of how to combine their complementary capabilities at inference time becomes critical. Yun et al. [1] identify a fundamental incompatibility between MDLMs and existing ensemble methods: autoregressive (AR) ensemble techniques (logit averaging, confidence-based routing) rely on a shared “next-token” concept defined by a fixed left-to-right generation order, but MDLMs denoise tokens in a flexible, confidence-determined order. At any given denoising step, two different MDLMs will generally have different tokens unmasked at different positions, making token-level or span-level fusion impossible.

The paper establishes two empirical observations that motivate their solution. First, *correct samples exhibit more stable confidence dynamics on answer tokens*: the Token Change Count (TCC) metric—counting how many times the top-1 predicted token at masked answer positions changes between consecutive denoising steps—is roughly $2\times$ higher for incorrect samples than correct ones (e.g., on MMLU: LLaDA correct $C = 1.81$ vs. incorrect $C = 4.27$; Dream correct $C = 2.32$ vs. incorrect $C = 6.19$). Second, *reliable partial trajectories from one model can correct another*: injecting as little as 33% of a correct model’s denoising trajectory into an incorrect model yields a 56–81% correction rate, demonstrating that cross-model trajectory sharing can rescue failing generations.

1.2 Method

Background: MDLM Forward/Reverse Process. The forward process corrupts clean tokens x into noisy states z_t via:

$$q(z_t | x) = \text{Cat}(z_t; \alpha_t \cdot x + (1 - \alpha_t) \cdot m) \quad (1)$$

where α_t is a noise schedule and m is the mask token. The reverse process is parameterized by a neural network f_θ that estimates the clean-token distribution at each masked position, trained by minimizing the negative ELBO over masked positions.

TIE: Trajectory-based Iterative Ensembling. TIE operates over M constituent MDLMs in a recurring three-step cycle:

Step 1—Trajectory Generation: Each model m independently decodes from its current state for n steps (the “ensemble interval”), producing a partial trajectory $T_m^{(n)}$. A critical design choice

enforces that answer-token positions are unmasked *only after* all reasoning-token positions are fully resolved, ensuring reasoning precedes answers.

Step 2—Trajectory Assessment: Four scoring metrics evaluate trajectory quality, all computed exclusively over masked answer-token positions $\mathcal{A}_m^{(t)}$:

(a) **Negative Token Change Count (history-based):**

$$\text{Score}(T_m^{(n)}) = -\tilde{C}_m^{(n)} = - \sum_{t=T-n+1}^{T-1} \frac{1}{|\mathcal{A}_m^{(t)}|} \sum_{a \in \mathcal{A}_m^{(t)}} \mathbf{1}[\arg \max p_a^{(t)} \neq \arg \max p_a^{(t+1)}] \quad (2)$$

The normalization by $|\mathcal{A}_m^{(t)}|$ is essential because models may have different numbers of masked answer positions at each step, preventing unfair penalization of models with more remaining masks.

(b–d) **Cross-model scoring for logit-based metrics** (top-1 probability, entropy, probability margin):

$$\text{Score}(T_m^{(n)}) = \frac{1}{M} \sum_{m'=1}^M f(T_m^{(n)}; m') \quad (3)$$

Each trajectory is forwarded through *all* constituent models and scored using averaged confidence. This cross-model evaluation corrects for calibration differences—a single model might be systematically overconfident, biasing source-model-only scoring.

Step 3—Trajectory Relay: The best trajectory is selected and broadcast to all models:

$$m^* = \arg \max_{m \in [M]} \text{Score}(T_m^{(n)}), \quad T_m^{(n)} \leftarrow T_{m^*}^{(n)} \quad \forall m \in [M] \quad (4)$$

After relay, trajectory scores are reset and each model resumes independent decoding for the next n steps. This exploits a key property of MDLMs: since any model can resume denoising from any intermediate partially-unmasked state, the entire sequence can be shared as a “checkpoint” between models.

Final Response Selection. After decoding terminates, the M candidate responses are ranked by lowest normalized TCC over the entire generation, with ties broken by highest top-1 probability.

Implementation Details. Semi-autoregressive generation with block size 16 and low-confidence remasking with greedy decoding. Default ensemble interval $n = 16$ (32 for TCC on complex tasks). All models execute in parallel on separate GPUs, so wall-clock latency is comparable to a single model.

1.3 Results

LLaDA + Dream Ensemble. TIE consistently outperforms both individual models and post-generation ensembling across diverse benchmarks. Key results: MMLU 67.55 (vs. 67.46 Dream, 61.23 LLaDA), ARC-C **89.16** (+2.47 over best individual), GSM8K **83.62** (+4.85 over best individual), MATH500 48.6 (matching Dream), MBPP **64.17** (+0.94 over Dream). The TCC scoring function is the most robust, achieving the best results on 4 of 8 benchmarks.

Dynamic Leadership. The model producing the highest-scoring trajectory changes dynamically (44.80% change rate on GSM8K, 44.43% on Countdown), confirming that no single model dominates—different models contribute most at different denoising stages.

Code-Specialized Models. TIE generalizes to domain-specific MDLMs (DreamCoder + DiffuCoder), improving MBPP from 75.88 to **76.58** with top-1 scoring.

Compatibility with Acceleration. TIE maintains improvements under both threshold-based unmasking ($\tau = 0.9$) and top- k unmasking ($k = 2$), confirming compatibility with existing decoding acceleration techniques.

1.4 Limitations

Large capability gaps degrade performance: When one model substantially underperforms (gap >15%), TIE can hurt rather than help. On HumanEval, where LLaDA (45.73) is far weaker than Dream (61.59), TIE underperforms Dream alone—the weaker model introduces noisy trajectory signals.

Limited to two-model ensembles: All experiments use $M = 2$. Scaling behavior with larger ensembles is unexplored.

Answer-position dependence: The framework requires clear delineation between “reasoning tokens” and “answer tokens” (via “Answer:” suffix), which may not generalize to open-ended generation tasks without structured output formats.

No learning component: While training-free is presented as a strength, TIE cannot learn to correct its own trajectory selection biases from past experience.

1.5 Relevance to Our Research

TIE addresses the MDLM ensembling problem by operating at the *trajectory level* rather than the token level—a natural choice given MDLMs’ flexible generation order. This connects directly to our extensive body of diffusion LLM reviews. Our DMax review [4] and Orthrus review [5] explored parallel decoding strategies that accelerate single-model MDLM inference; TIE complements these by showing how to combine *multiple* models during the same denoising process. The trajectory relay mechanism—where one model’s partially-denoised sequence serves as a “checkpoint” for all models—is conceptually similar to the remasking strategies analyzed in S2D2 [6], but applied across models rather than within a single model’s speculative decoding loop.

The TCC metric (stability of answer-token predictions across denoising steps) provides an empirical quality signal complementary to the D²-Monitor’s [7] hesitation-aware routing, which detects safety-critical positions by measuring prediction entropy spikes. TIE’s cross-model scoring (Eq. 3) also relates to our TIDE review [8], where cross-architecture distillation transfers knowledge between AR and diffusion LLMs—here, cross-model evaluation during inference achieves knowledge fusion without any training.

The observation that injecting just 33% of a correct trajectory suffices for 56–81% correction rate has strong implications for on-policy distillation: in the geometric OPD framework [9], we found that the student’s update subspace locks early in training. TIE suggests that even partial teacher trajectories at critical denoising steps could break these locks, motivating *trajectory-level* (not just token-level) teacher intervention in OPD.

2 Follow-up Research Directions

2.1 Direction 1: Trajectory-Guided On-Policy Distillation for MDLMs

Gap. Current OPD methods for diffusion LLMs (OPDLM [10], geometric OPD [9], FiRe-OPD [11]) operate at the token level: teacher supervision is applied uniformly or with per-token reweighting across all denoising steps. TIE’s 56–81% trajectory correction finding suggests that

which denoising steps receive teacher supervision matters far more than how much supervision each token gets. Yet no OPD method for MDLMs exploits trajectory-level teacher intervention.

Approach. Adapt TIE’s assessment-relay mechanism to OPD: during student denoising, evaluate partial trajectories using TCC computed against teacher predictions (rather than cross-model scoring). At denoising steps where student TCC exceeds a threshold (indicating unstable, likely-incorrect token predictions), inject the teacher’s partial trajectory via relay and let the student resume from the teacher’s state. This creates an *adaptive trajectory injection* scheme where teacher compute is allocated to denoising steps where the student struggles most. Key distinction from uniform OPD: the teacher only needs to run forward passes at high-TCC steps (estimated 20–30% of total steps based on TIE’s finding that trajectory changes cluster at specific phases), reducing teacher inference cost by 3–5 $\times$ while concentrating supervision where it matters.

Feasibility. TCC computation reuses student logits already available during denoising—no additional forward passes. Teacher trajectory generation at high-TCC steps uses standard MDLM sampling. Validation on LLaDA-8B (student) / LLaDA-70B (teacher) with MMLU and GSM8K, directly comparable to OPDLM’s experimental setup. The $n = 16$ ensemble interval from TIE provides a validated default for the injection frequency.

2.2 Direction 2: Self-Improving MDLM Ensembles via Learned Trajectory Selection

Gap. TIE uses fixed, hand-designed scoring functions (TCC, top-1 probability, entropy, margin) for trajectory assessment, with no mechanism to learn which scoring function works best for which task or denoising phase. Table 3 of the paper shows significant scoring function sensitivity—TCC wins on 4/8 benchmarks, entropy on 3/8, top-1 on 1/8—but TIE cannot adapt its selection strategy from experience. This connects to the skill curation challenge identified in our SkillOS [12] and SkillOpt [13] reviews, where agents learn to select and compose skills over time.

Approach. Replace TIE’s fixed scoring with a lightweight *meta-selector* trained to predict which scoring function (or convex combination thereof) will yield the best trajectory at each relay step. Input features: current denoising step t/T , number of remaining masked tokens, cross-model agreement (fraction of answer positions where both models predict the same top-1 token), and the four scoring metrics themselves. The meta-selector is a small MLP (<1M parameters) trained on a held-out set of tasks where ground-truth correctness is available, using a simple binary reward (did the selected trajectory lead to a correct final answer?). This learns task-phase-dependent scoring preferences—e.g., preferring entropy-based scoring at early denoising steps (high uncertainty) and TCC at later steps (stability matters more). The meta-selector can be updated continually as new tasks are encountered, creating a *self-improving* ensemble that accumulates trajectory selection experience, addressing the continual learning dimension of our research [14].

Feasibility. The meta-selector is extremely lightweight (forward pass <0.1ms, negligible compared to model inference). Training data is generated automatically from TIE runs with known ground truth. The four scoring metrics are already computed in TIE, so the meta-selector only adds feature extraction and a single MLP forward pass per relay step. Validation: compare meta-selector TIE vs. fixed-scoring TIE on the same benchmark suite, measuring both accuracy and the learned scoring preferences across denoising phases.

2.3 Direction 3: Divergence-Aware Trajectory Relay for Flow-Based Diffusion Models

Gap. TIE is designed specifically for *masked* diffusion (discrete state space with mask tokens), but the rapidly growing family of *flow-based* diffusion models (LangFlow [15], Flow-DPPO [16], Flow-OPD [17]) operates in continuous latent spaces where the “trajectory” concept is fundamentally different. In flow matching, denoising follows ODE/SDE paths in continuous space rather than discrete unmasking sequences. No ensemble method exists for combining multiple flow-based diffusion LLMs during inference.

Approach. Extend TIE to continuous flow matching by redefining trajectory assessment in terms of *velocity field divergence*. At each ODE step t , compute the cross-model velocity disagreement: $D_t = \|v_{\theta_1}(z_t, t) - v_{\theta_2}(z_t, t)\|_2$, where v_{θ_i} are the velocity fields of two flow models. High D_t indicates the models disagree on the denoising direction—analogous to high TCC in masked diffusion. For trajectory relay: at each ensemble interval, project both models’ intermediate latents $z_t^{(1)}, z_t^{(2)}$ to token space via the nearest-neighbor decoder, score using cross-model log-likelihood (each model evaluates the other’s projected tokens), and relay the higher-scoring latent to both models. The key technical challenge is that flow models operate in continuous space, so “relay” requires mapping between potentially different latent representations. Our approach uses the shared vocabulary decoder as a common projection space, inspired by TIDE’s cross-architecture distillation [8]. This would also leverage Flow-DPPO’s [16] ratio-noise decomposition ($\text{Var}[\log r_t] = 2D_{\text{KL}}$) as an alternative trajectory quality metric—trajectories with lower accumulated KL drift are more reliable.

Feasibility. Velocity field evaluation reuses model forward passes already performed during ODE solving. The nearest-neighbor decoding projection is a single matrix multiply. Main risk: continuous-space relay may introduce discretization artifacts when projecting to token space and back. Validation on LangFlow-7B variants with different training seeds (creating heterogeneous models), measuring perplexity and downstream task accuracy on LAMBADA and HellaSwag.

3 Conclusion

TIE introduces the first ensembling framework specifically designed for Masked Diffusion Language Models, addressing the fundamental incompatibility between MDLMs’ flexible generation order and existing AR-based ensemble methods. The three-step cycle—*independent trajectory generation*, *cross-model trajectory assessment* (via TCC, top-1 probability, entropy, or margin scoring), and *trajectory relay*—exploits MDLMs’ unique property that any model can resume denoising from any partially-unmasked state. The empirical finding that correct generations exhibit $2\times$ lower token change count than incorrect ones provides a principled basis for trajectory quality assessment, while the 56–81% correction rate from partial trajectory injection demonstrates the power of cross-model knowledge sharing during inference. For our research program, TIE establishes trajectory-level fusion as the natural ensembling granularity for diffusion LLMs, complementing our prior work on token-level OPD (OPDLM, geometric OPD), step-level RL (V-GRPO, Flow-DPPO), and architecture-level distillation (TIDE). The trajectory relay mechanism suggests a new paradigm for on-policy distillation where teacher intervention operates at the trajectory checkpoint level rather than per-token supervision, potentially unifying the efficiency gains of selective distillation with the corrective power of full trajectory sharing.

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